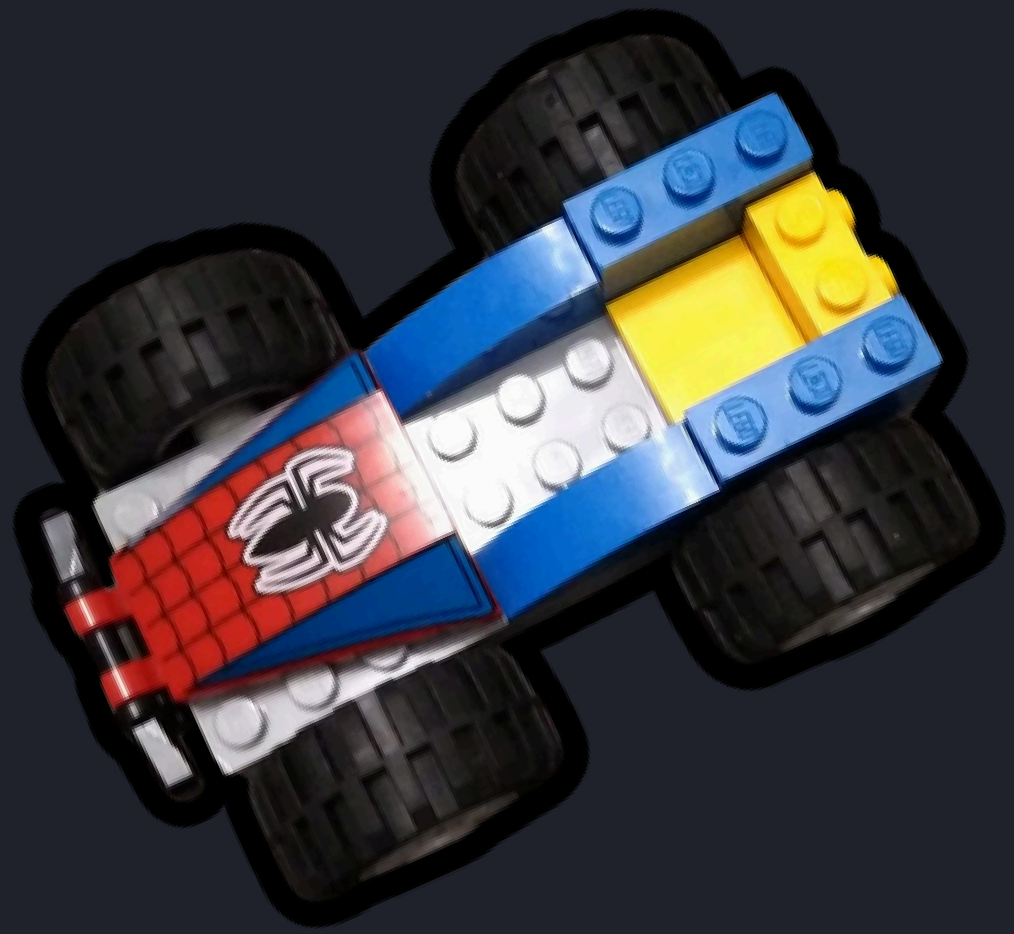
NEURO DRIVE

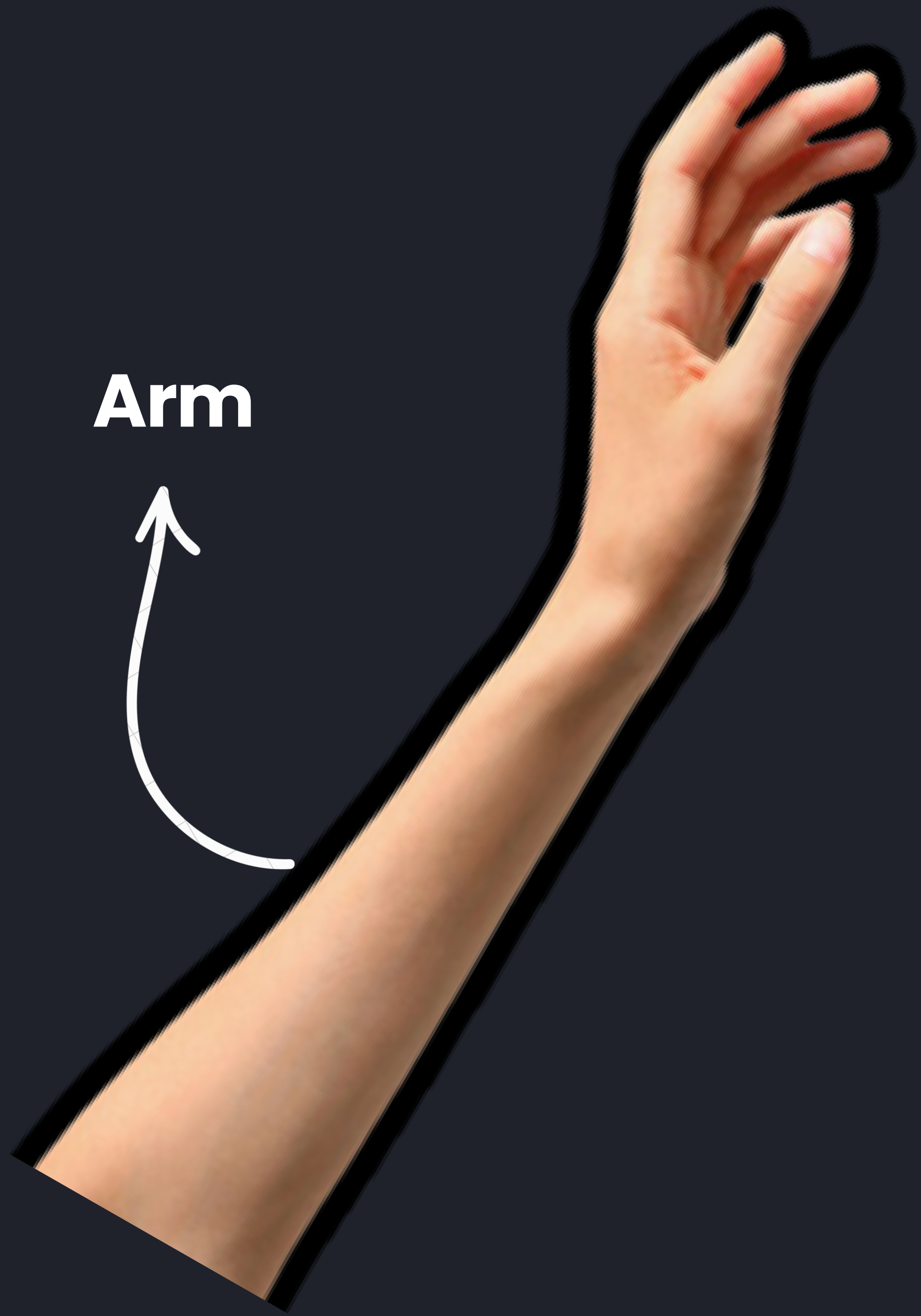

5

The World's Most Revolutionary, Groundbreaking, Unprecedented, Never-Before-Seen, Absolutely Incredible Robot Car

EMG web electrodess



Robot Car



Arm

MEET OUR TEAM

Kashvi Mummaneni

Aaliya Thakker

Abhay Gupta

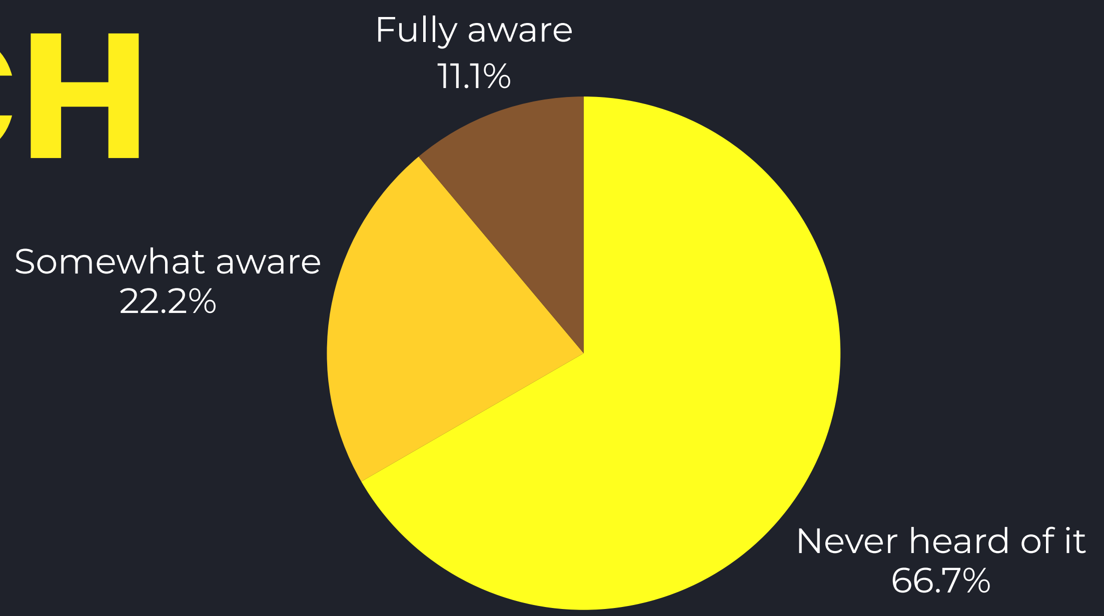
Reyansh Bhimrajka

THE PROBLEM

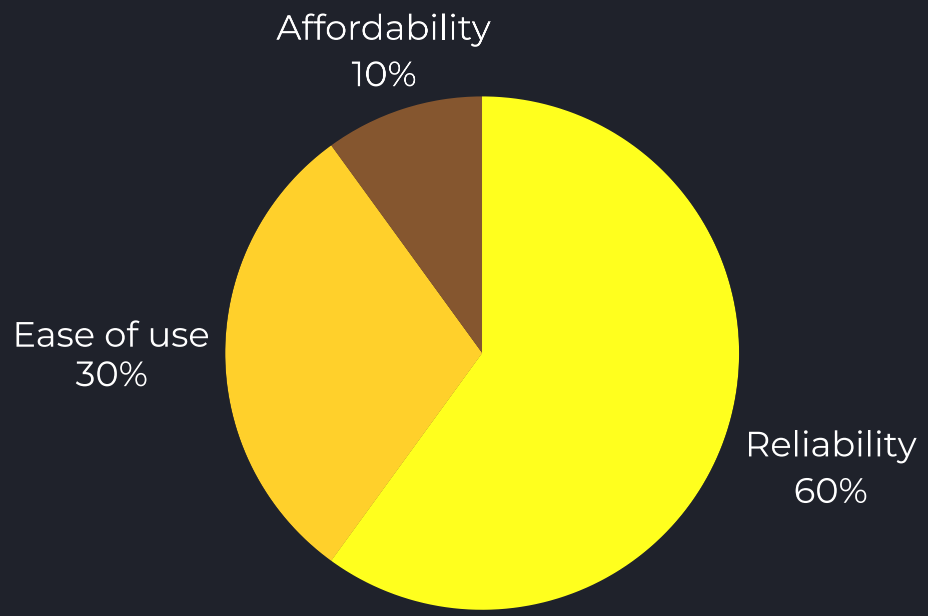
Everyday electronic devices like phones, remotes, controllers, robots; **require hands, fingers, and precise motor control to operate.** For people with conditions like paralysis, muscular dystrophy, ALS, or limb loss, this **creates a complete barrier to independent interaction with technology.**

However, most assistive technologies are either too expensive, too complex, or require full physical movement to operate.

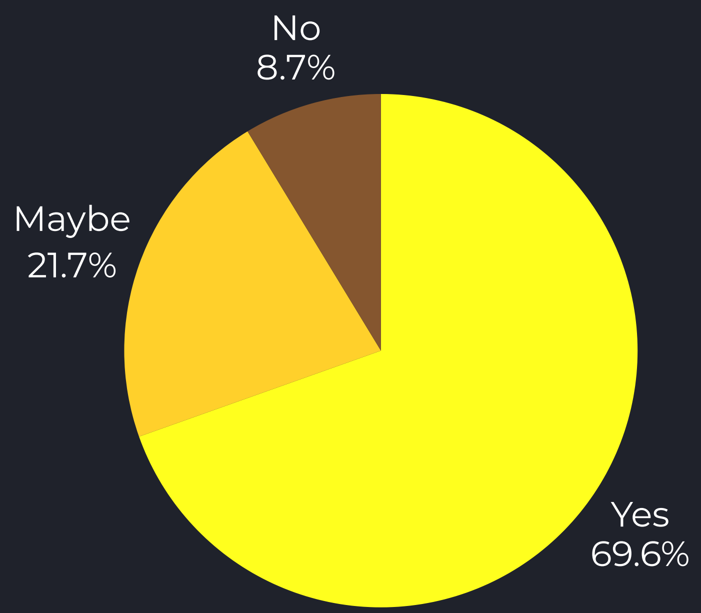
PRIMARY RESEARCH



The majority of respondents (66.7%) had never heard of EMG technology before. This highlights a clear gap in public awareness and reinforces the need for projects like NeuroDrive to make this technology more visible and accessible. The small percentage already familiar with it suggests strong potential for growth in this space.



Over half of respondents prioritized reliability above all else. This directly shaped our development approach — ensuring the EMG signal detection is consistent and accurate before adding complexity. Ease of use and affordability remain secondary considerations that we continue to work toward.



With 75% of respondents willing to use or recommend NeuroDrive, there is strong validated interest in hands-free muscle-controlled devices. This gives us confidence that the concept resonates beyond the engineering space and has real-world appeal among general users.

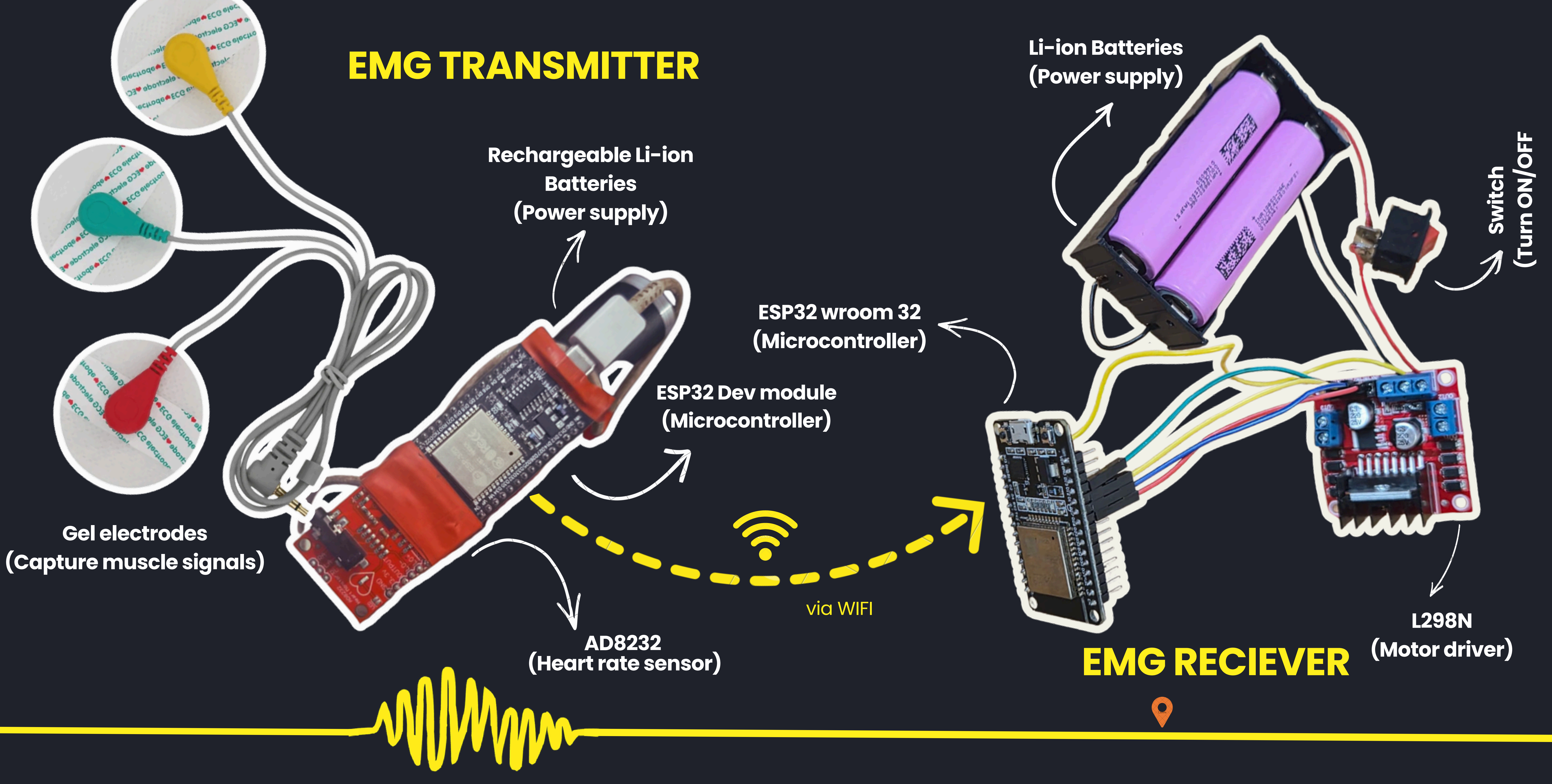
NeuroDrive is a hands-free robot car controlled entirely by **muscle signals.** Using **EMG sensors** placed on the arm, it detects the electrical activity produced by **muscle contractions** and translates them into **movement commands.**

CLOCKWISE ROTATION

FORWARD MOVEMENT


make.
break.
create!

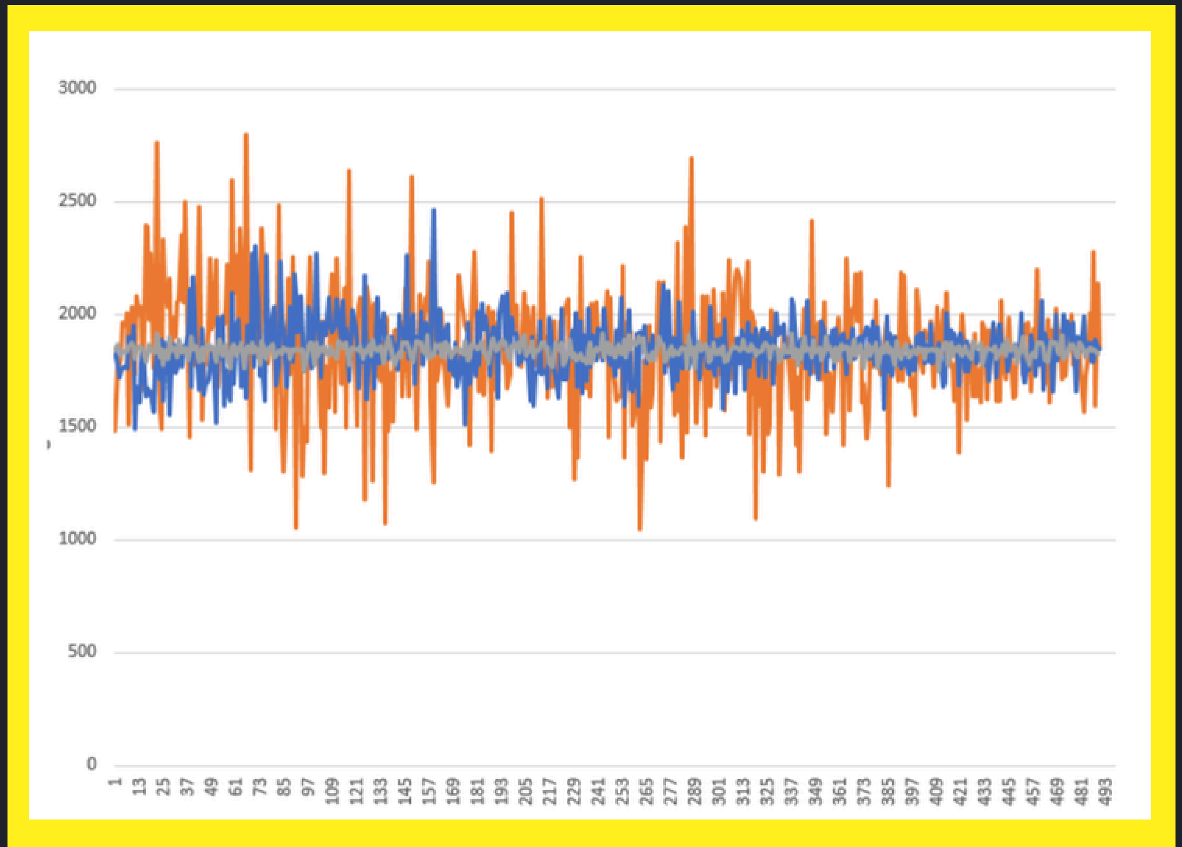
HARDWARE BREAKDOWN



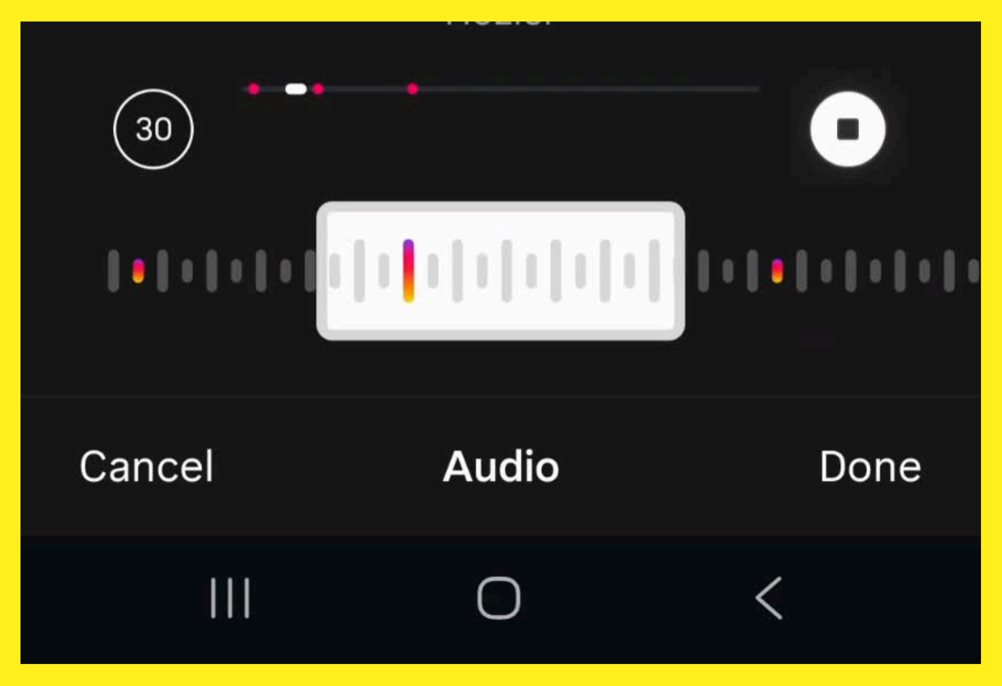
TECHNICAL OVERVIEW OF HOW **NEURO DRIVE** WORKS

The receiver ESP32 creates a **WiFi hotspot** ("EMG_CAR"). The transmitter connects to it and **sends single-character** commands (F, L, or S) as UDP packets to IP 192.168.4.1.

Every 100ms, **a new EMG reading enters a 50-sample array, replacing the oldest.** The window always holds exactly 5 seconds of signal history, continuously updated in real time.



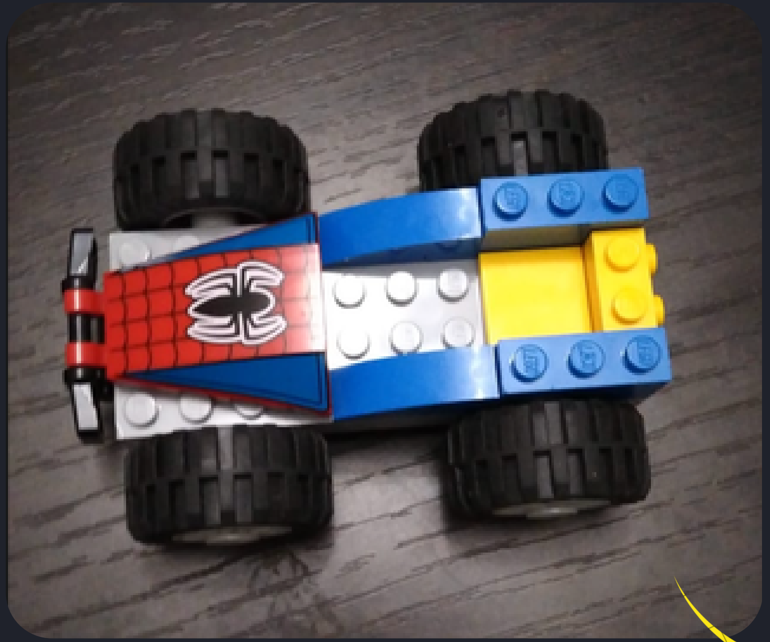
State	Mean	Min	Max	Amplitude (Range)
Relaxed	1845	1747	1919	172
Weak flex	1860	1495	2468	973
Strong flex	1865	1051	2799	1748



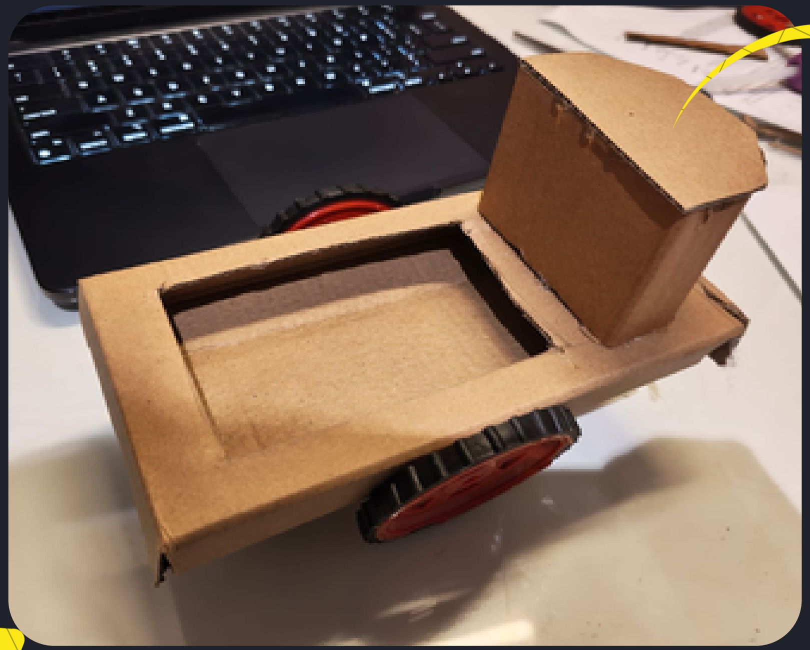
State	Amplitude Range
Relaxed	0 – 200
Weak flex	900 – 1000
Strong flex	1700 – 1800

DESIGN PROCESS

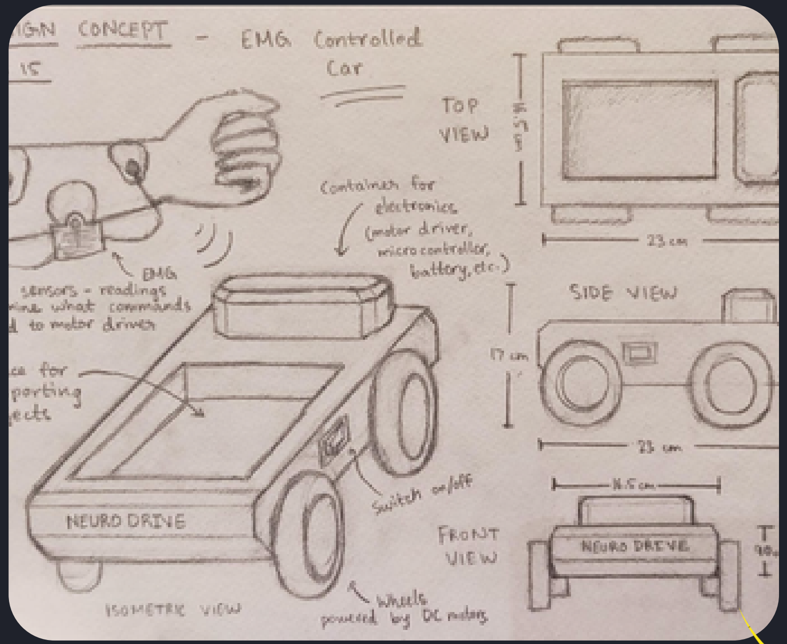
Over the past **three months**, we developed our project from an initial idea into a **working prototype.**
Throughout the process, we conducted **research**, learned the necessary **skills**, planned and **designed** the product, and finally built and tested it here at the **residency**.
The following slides show our **journey** and the steps we took along the way.



PAPER PROTOTYPE 1

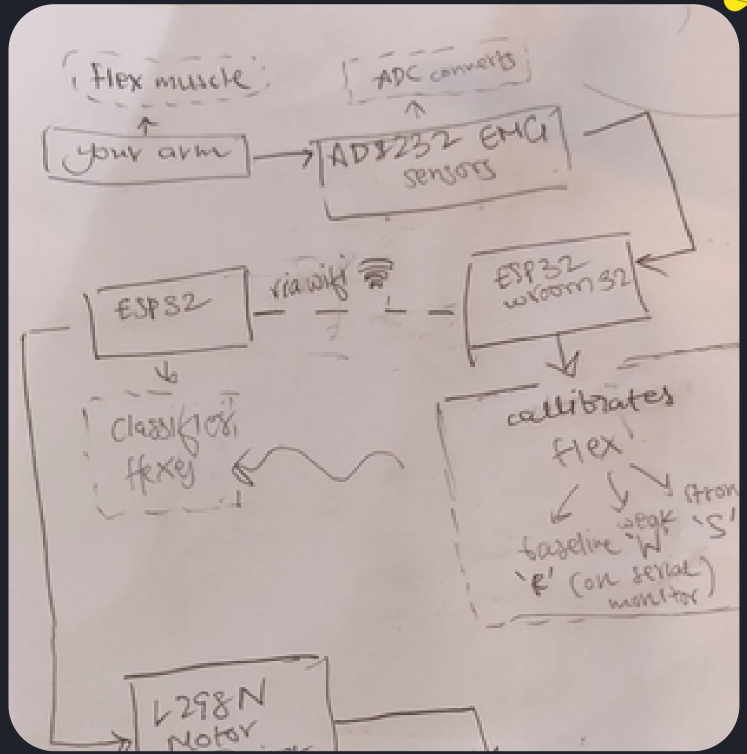


PAPER PROTOTYPE 2



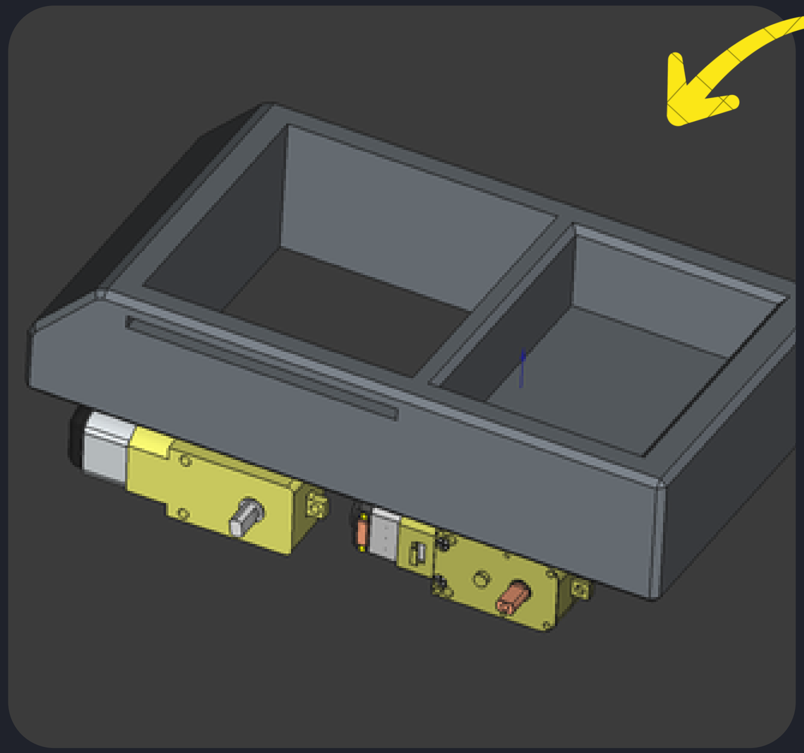
DESIGN SKETCH

COMPONENTS

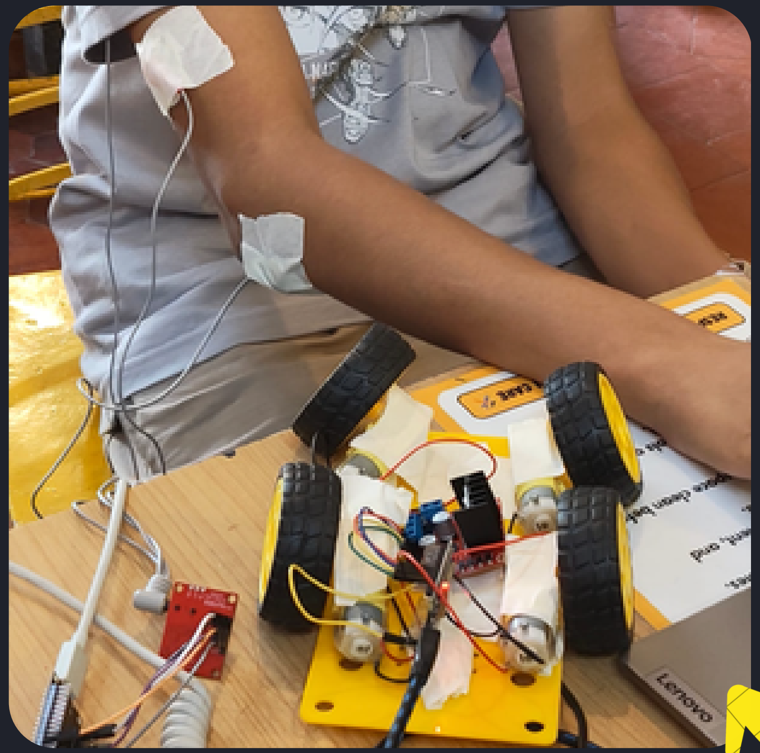


BLOCK DIAGRAM

ELECTRONICS & CIRCUIT

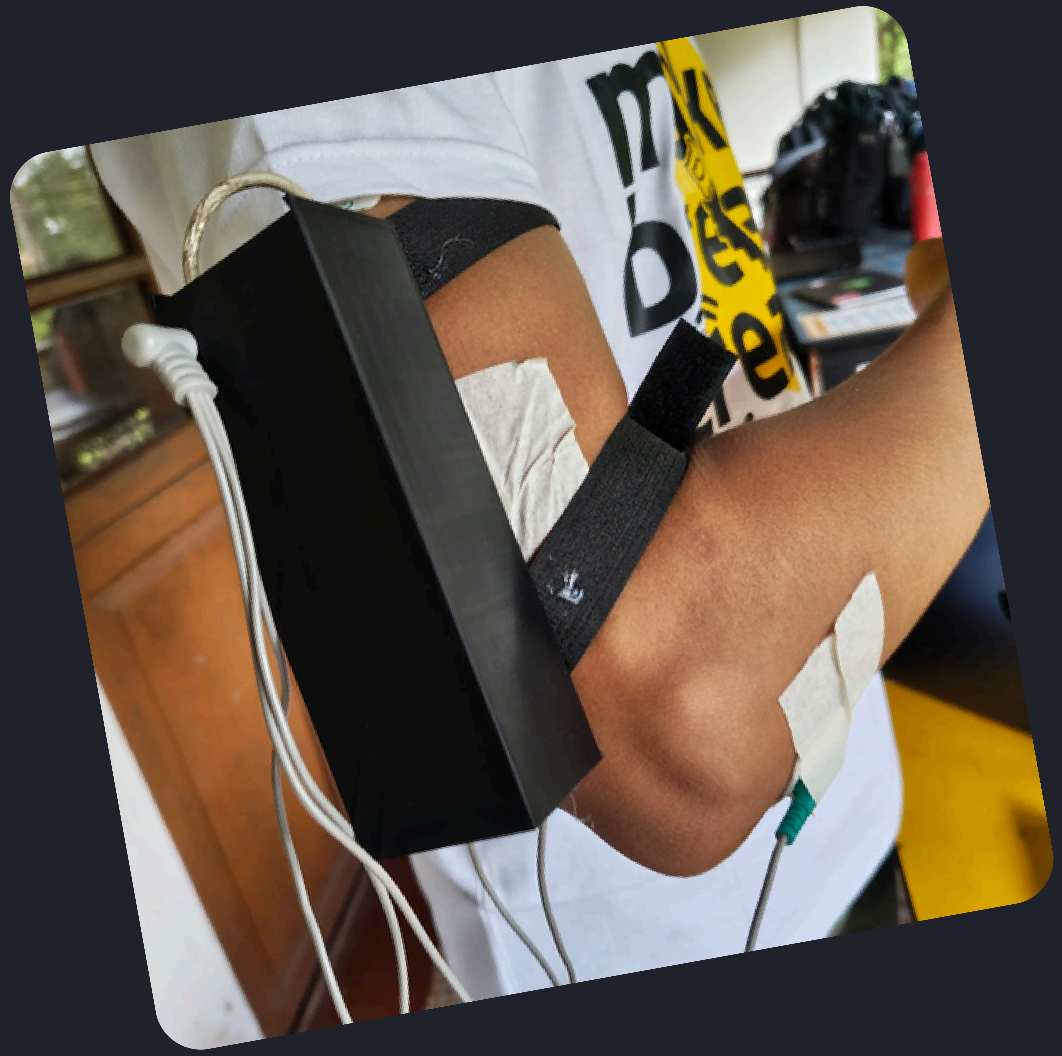


CAD



TESTING (AGAIN & AGAIN)

ECG SENSOR



FINAL PRODUCT

ROBOT CAR